

Automation System Design

Task 3 — Smart Home Lighting: Flowchart and Explanation Report

1. System Overview

This report presents the design of an automated smart home lighting system. The system's goal is to turn lights on automatically when a room is occupied and dark, and to turn them off automatically when the room is empty, while still allowing the occupant to override the automatic behavior at any time using a wall switch or smartphone app. This design mirrors the logic used in real commercial smart lighting products, combining two sensor inputs with a timeout-based control loop.

2. Components

- PIR (Passive Infrared) Motion Sensor — detects the presence of movement in the room, used as the primary trigger for occupancy.
- LDR (Light Dependent Resistor) / Ambient Light Sensor — measures the current brightness of the room, so lights are not turned on unnecessarily during daylight.
- Microcontroller (e.g., Arduino or ESP32) — reads both sensors, runs the control logic described in the flowchart, and drives the relay.
- Relay Module — an electrically controlled switch that allows the low-voltage microcontroller to safely switch the mains-voltage lighting circuit on and off.
- LED Light Fixture — the load being controlled.
- Manual Override Input (wall switch or smartphone app via Wi-Fi/Bluetooth module) — allows the user to force the lights on or off regardless of what the sensors report.

3. System Flowchart

The diagram below shows the full control loop of the system, including the sensor-driven automatic behavior and the manual override path.

Smart Home Lighting — Automation Flowchart

Motion + Ambient-Light Control Loop with Manual Override

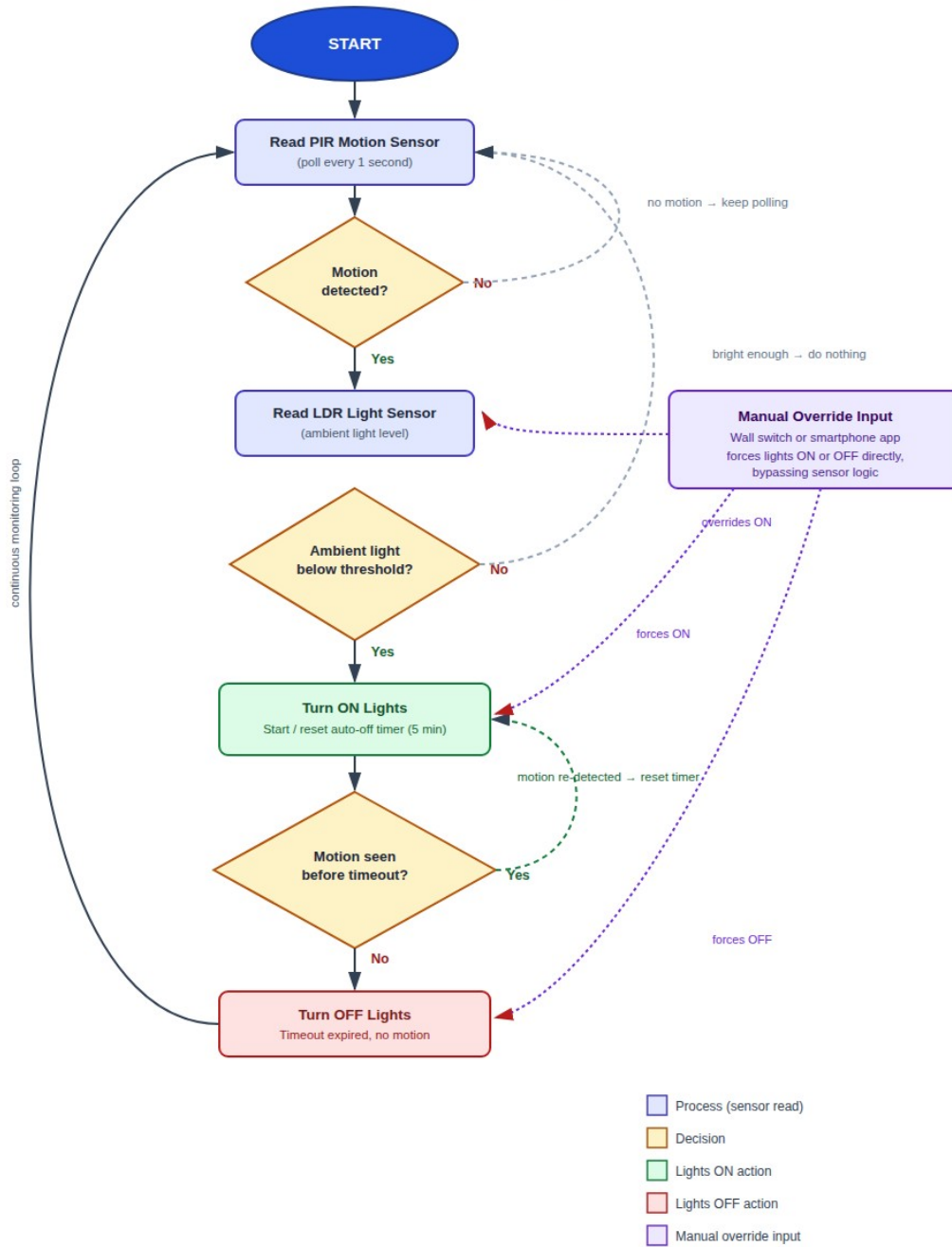


Figure 1 — Smart home lighting automation flowchart: sensor polling, decision logic, timeout-based auto-off, and manual override.

4. Logic Explanation

The system continuously polls the PIR motion sensor roughly once per second. If no motion is detected, it simply loops back and keeps polling, consuming minimal power while idle. Once motion is detected, the system checks the LDR ambient light sensor. If the room is already bright enough (for example, during daytime), the system takes no action and returns to polling, avoiding unnecessary energy use. If the room is dark and motion is present, the system turns the lights on and starts a five-minute auto-off timer.

While the lights are on, the system continues monitoring for motion. Each time new motion is detected before the timer expires, the timer resets, keeping the lights on for as long as the room remains occupied. If no motion is detected for the full timeout period, the system automatically turns the lights off and returns to the idle polling state, closing the loop.

The manual override path runs independently of this sensor logic. At any point in the cycle, a wall switch or smartphone app command can force the lights on or off directly. This is essential in practice, since sensor-only automation can occasionally misread a room as empty (for example, if a person is sitting very still) or the user may simply want a different lighting state than the sensors would choose, such as keeping a light on for a guest.

5. Benefits and Design Considerations

- Energy efficiency: lights are only powered when a room is both dark and occupied, and they turn off automatically when left on by mistake.
- Convenience: no manual switching is required for normal day-to-day use, since the system reacts automatically to occupancy.
- Reliability through override: the manual control path ensures the user is never fully dependent on sensor accuracy.
- Extensibility: this design can be extended with scheduling (e.g., disable automatic triggering late at night), multi-zone control for different rooms, or integration with a broader home automation hub.

6. Conclusion

This smart home lighting system demonstrates how a small number of low-cost sensors, combined with simple conditional logic and a timeout mechanism, can automate a common household task while remaining energy-efficient and fully overridable by the user. The same fundamental pattern — sensor input, decision logic, timed action, and manual override — is broadly applicable to many other automation systems, including HVAC control, security lighting, and irrigation systems.